Python Modules

What is a Module?

- A **module** in Python is simply a **Python file (.py)** that contains functions, classes, or variables.
- Modules help in **organizing code** and **reusing it** across multiple programs.
- Example: If you have math.py with functions, you can use them in another file with import math.

✅ **Example 1: Using Functions from a Module**

📂 **Files:**

operations.py

```
def add(num1, num2):
    print(num1 + num2)
def mul(num1, num2):
    print(num1 * num2)
```

client.py

```
# Approach 1: Import whole module
    import operations
    operations.add(10, 20)
    operations.mul(10, 20)
# Approach 2: Import specific functions
    from operations import add, mul
    add(10, 20)
    mul(10, 20)
# Approach 3: Import everything
    from operations import *
    add(10, 20)
    mul(10, 20)
```

Key Points:

- import module_name → use with module_name.function().
- from module_name import func1, func2 → directly call the functions.
- from module_name import * → imports all functions, but may cause conflicts if two modules have functions with the same name.

✅ **Example 2: Same Function Names in Two Modules**

📂 **Files:**

animal.py

```
def fly():
    print("Animal Cant fly")
def color():
    print("Animal is Black")
```

bird.py

```
def fly():
    print("Bird Can Fly")
def color():
    print("Bird is Green")
```

main.py

```
# Approach 1: Import modules separately
    import animal
    import bird

    animal.fly()
    animal.color()

    bird.fly()
    bird.color()
```

```
# Approach 2: Import everything from both
    from animal import *
    fly()
    color()

    from bird import *
    fly()
    color()
```

Key Points:

- If two modules have functions with the **same name**, importing with from module import * may cause confusion.
- To avoid conflicts, prefer **approach 1** (use module_name.function()).

✅ **Example 3: Classes in Different Modules**

📂 **Files:**

a.py

```
class Animal:
  def display(self):
    print("I like Cow")
```

b.py

```
class Bird:
  def display(self):
    print("I like Parrot")
```

main.py

```
# Approach 1: Import modules
    import a
    import b
    obj1 = a.Animal()
    obj1.display()
```

```
obj2 = b.Bird()
obj2.display()
```

Approach 2: Import classes directly

```
from a import Animal
from b import Bird

obj3 = Animal()
obj3.display()

obj4 = Bird()
obj4.display()
```

Key Points:

- You can import entire modules or specific classes.
- If only one or two classes are needed, **direct import is cleaner**.
- If many functions/classes are needed, import the full module.